higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# NATIONAL CERTIFICATE (VOCATIONAL)

## SYSTEM ANALYSIS AND DESIGN
### NQF LEVEL 4

(10041004)

**1 March 2019 (Y-Paper)**
**13:00–16:00**

**This question paper consists of 13 pages.**

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**TIME: 3 HOURS**
**MARKS: 200**

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**INSTRUCTIONS AND INFORMATION**

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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## SECTION A

## QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10) in the ANSWER BOOK.

1.1.1 A term often used to describe individuals' anonymity and how safe they feel when sharing their digital information:

A Welfare
B Privacy
C Protection
D Safety

1.1.2 The politically motivated use of computers and information technology to cause severe disruption or widespread fear is an example of … behaviour.

A unethical
B unprofessional
C unaccountable
D unskilled

1.1.3 To minimise risk, a systems analyst has to …

(i) identify potential risks and threats and mitigate them
(ii) identify, assess, control and review risks
(iii) avoid, reduce, transfer and accept risks

Which of the following is correct:

A (i) and (ii)
B Only (ii)
C (ii) and (iii)
D (i) and (ii) and (iii)

1.1.4 Which ONE of the following information-gathering techniques would be best suited for people with language barriers:

A Interview
B Group interview
C Questionnaire
D Observation

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1.1.5 Selecting the number or size of the data to examine, or people to interview, question or observe, to gather data is called:

A sampling
B data gathering
C comparing
D studying

1.1.6 … is a methodology that involves the client or end user in the design and development of an application.

A rapid application development
B agile development
C spiral development
D joint application development

1.1.7 A developer can use the following controls to select only ONE particular item when presented with more than one item:

(i)  CheckedListBox
(ii)  ComboBox
(iii)  ListBox
(iv)  TextBox

Which of the following is correct:

A (i) and (ii)
B Only (iv)
C Only (ii)
D (i), (ii) and (iii)

1.1.8 This type of testing occurs in the final phases of testing before the information system goes live:

A static testing
B user-acceptance testing
C dynamic testing
D alpha testing

1.1.9 An advantage of … implementation is that it provides a backup in case the new system fails.

A immediate
B phased
C parallel
D pilot

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1.1.10 … is a system of hardware and/or software patterned after the operation of the human brain.

A Neural network
B Natural language processing
C Synapse
D Virtual reality

(10 × 1) (10)

1.2 Choose the correct word or words from those given in brackets. Write only the word or words next to the question number (1.2.1–1.2.10) in the ANSWER BOOK.

1.2.1 Virtual (reality/coding) is an interactive computer-generated experience taking place within a simulated environment.

1.2.2 Creating a working (case/prototype) is another way of simulating a system in the user's workplace when testing a design.

1.2.3 The (duration/availability) of a system is the probability that a system is functioning when needed under normal conditions.

1.2.4 A (checksum/hash) is an extra digit representing the sum of all the correct digits.

1.2.5 A (simulation/model) is a visual presentation of a system that allows one to show, diagrammatically, what a system does and how it does it.

1.2.6 The primary aim of the (observation/interview) method of data gathering is to record how the user interacts with the system.

1.2.7 An interviewee can answer (open/closed) questions in as much detail as he/she can.

1.2.8 (Scope-creep/Consequence) is a word used to describe the size of the outcome if a risk was realised.

1.2.9 A/An (firewall/antivirus) is a type of software that monitors incoming and outgoing network traffic and decides whether to allow or block specific traffic based on a defined set of rules.

1.2.10 A (management/professional) body sets up standards of training and ethics to determine if people work professionally.

(10 × 1) (10)

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1.3 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.3.1–1.3.10) in the ANSWER BOOK.

1.3.1 One of the advantages of using fuzzy logic is that its language is easier to read and understand than formal logic.

1.3.2 Checking whether the user has met all objectives and satisfied all user requirements is a reason why one should review the system after implementation.

1.3.3 Verification and validation falls under the aspect of designing a new system.

1.3.4 Alpha testing occurs when a tester tests the inner working part of the information system.

1.3.5 Two methods of documenting process flows include the use of use-cases and data flow diagrams.

1.3.6 It is important for the interviewer to discuss the objectives of the interview with the interviewee to ensure that he/she understands him/her properly.

1.3.7 A data dictionary can be created to enable all project stakeholders to understand what data the system will use and maintain.

1.3.8 Regular backups of software on a disk are an example of software piracy.

1.3.9 Malware is any software intentionally designed to cause damage to a computer, server or computer network.

1.3.10 A dialog box is a small area on-screen in which the user is prompted to provide information or confirm particular operations.

(10 × 1) (10)

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1.4 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–S) next to the question number (1.4.1–1.4.10) in the ANSWER BOOK.

| COLUMN A | | COLUMN B | |
|---|---|---|---|
| 1.4.1 | The right to be treated without discrimination, especially on the grounds of one's sex, race, or age | A | risk assessment |
| 1.4.2 | Minimising, checking and controlling the probability of unfortunate occurrences | B | close ended |
| 1.4.3 | Adding features and functionality without due approval | C | validation |
| 1.4.4 | Person supplying information through a questionnaire | D | activation function |
| 1.4.5 | Questions which can be answered by a simple "yes" or "no" | E | equal opportunity |
| 1.4.6 | Types include level 1 and context | F | check boxes |
| 1.4.7 | Storage of information without requiring a network connection | G | open questions |
| 1.4.8 | Set of actions used to test for compliance in certain respects | H | scope-creep |
| 1.4.9 | A component of a neural network | I | respondent |
| 1.4.10 | Occurs prior to the client accepting the new system | J | data flow diagram |
| | | K | synapses |
| | | L | off-line |
| | | M | beta test |
| | | N | access control |
| | | O | permission |
| | | P | interviewee |
| | | Q | process flow diagram |
| | | R | empowerment |
| | | S | open ended |

(10 × 1) (10)
**[40]**

**TOTAL SECTION A: 40**

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## SECTION B

## QUESTION 2

Read the snippet below and answer the questions.

2.1

> In addressing copyright infringement of software in South Africa and around the world. The Software Alliance (formerly the Business Software Alliance - BSA) has been focussing more on policy and education than enforcement.
>
> 'We need to look at the reasons people pirate and address them, rather than just enforcing antipiracy laws,' added Monique Ferreira from Microsoft South Africa's antipiracy and software asset-management division.

[Adapted from: *https://mybroadband.co.za/vb/showthread.php/608430-Pirate-software-costs-SA-business-millions-in-fines-BSA*]

2.1.1 Do you think the move by the BSA to focus more on educating people about piracy is more effective than enforcing antipiracy laws? (Answer Yes or No) (1)

2.1.2 Motivate your answer to QUESTION 2.1.1. (2)

2.1.3 Recommend FOUR strategies, other than education, to the BSA to help them reduce software piracy. (4)

2.1.4 List TWO ways in which software piracy affects the South African ICT industry. (2)

2.2 Define the term *hard-disk loading*. (1)

2.3 Explain the term *code of practice*. (2)

2.4 There are many professional bodies seeking to improve the computer industry worldwide.

Name any TWO professional bodies that relate only to the South African ICT industry. (2)

**[14]**

## QUESTION 3

3.1 Define the term *availability of information*. (1)

3.2 Explain ONE difference and ONE similarity between the terms *confidentiality* and *integrity of information*. (2 + 2) (4)

3.3 List the SEVEN basic steps in a full risk assessment. (7)

3.4 Define the term *system engineering*. (1)

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3.5 List any TWO of the three deliverables of the system requirement specification (SRS) document. (2 × 1) (2)

3.6 Explain the term *scope of the project*. (2)

3.7 List FOUR criteria to determine if a project or system is a success. (4)

3.8 Read the following snippet and answer the questions.

| |
|---|
| You have been appointed as a security analyst for the TVET college where you studied. Your first task is to analyse the current state of security of the colleges servers based in the server room. |

3.8.1 Design a security checklist with THREE criteria to assess the security of the servers.

Use the following as a guide:

| Checklist item | |
|---|---|
| 1. E.g.: Servers are … | ☐ |

**NOTE:** The security items must be related to the servers only.
(3 × 1) (3)

3.8.2 Suggest FOUR measures for increasing the security on the workstations used at the college. (4)

3.9 Define the term *encryption*. (1)
**[29]**

## QUESTION 4

Read the snippet below and answer the questions.

4.1

| |
|---|
| Business Enterprise Solutions (BES) is an IT company that specialises in developing computer systems for different organisations. BES specialists must first gather data and the necessary information that would help them when developing these systems. The systems must also be able to solve the organisation's problems and fulfil its needs. The senior systems analyst has decided to conduct interviews, ask different questions to get detailed answers and observe the employees in the workplace to help them during the development stages. Currently BES is tasked with the design and development of a new college hostel administration system which will replace the older system. Some analysts are busy with interviews whilst others are reviewing the existing systems documentation. |

4.1.1 Explain how the BES system analysts can ensure that the interviewee understood the questions asked. (3)

4.1.2 The senior systems analyst has instructed the team to NOT use any closed-ended questions when conducting the interview.

Give TWO reasons why the senior analyst has decided against the use of closed-ended questions. (2 × 2) (4)

4.1.3 Differentiate between a *one-on-one interview* and a *group interview.* (2)

4.1.4 Give ONE advantage of conducting a one-on-one interview. (2)

4.1.5 Formulate any THREE questions that the system analysts can ask the users of the old system as part of an observation (as a data gathering process). (3)

4.1.6 After having gathered all the data using the interview and observation, the senior systems analyst has instructed the team to develop a questionnaire to gauge the satisfaction levels of the users regarding their experience of interacting with BES analysts and their perceptions of the new system.

Design and formulate a questionnaire with THREE questions.
Use a different type of question for each of the questions you chose. Indicate the type of question, e.g. Likert Scale for each question as well. (3 × 2) (6)

4.2 Name TWO similarities between the interview and questionnaire methods of data gathering. (2)

4.3 List THREE types of descriptions a data dictionary can contain. (3)

4.4 Define the term *composite key.* (2)

4.5 Define the term *bias*. (1)
**[28]**

**QUESTION 5**

Read the following scenario below and answer the question.

5.1

| |
|---|
| At the ABC Pharmacy a Pharmacist captures the prescription details of customers and dispenses the applicable medication. The pharmacist then has an invoice generated and assists with the processing of the payment. The Manager of the pharmacy checks the stock levels on a daily basis and reorders supplies and medicine where necessary. |

Use the given information to draw a use-case diagram for the scenario. Indicate ALL related actors, use cases, relationships and boundaries. (12)

Read the following scenario below and answer the question.

5.2 At the ABC Pharmacy a customer's prescription details, medical aid details and payment details are supplied. The pharmacist enters the prescription data and the system presents a list of generic alternatives. For each prescription label information is also supplied to the pharmacist. The manager of the pharmacy may supply the order details of new stock or obtain reports on the current stock levels.

Use the given information to draw an applicable DFD Level 0 (Context diagram). (13)

5.3 State THREE functions of a systems analyst. (3)

5.4 Differentiate between a *process flow diagram* (flow chart) and a *data flow diagram.* (3)

**[31]**

**QUESTION 6**

6.1 List THREE properties of identification codes. (3)

6.2 State TWO limitations of text-based output. (2)

6.3 Give ONE common example of a typical text-based report generated by a POS system. (1)

6.4 When creating a design for inputs and outputs, a systems analyst must avoid errors in the interface design.

Differentiate between avoiding errors in *online input* and *off-line input*. (3 + 3) (6)

**[12]**

**QUESTION 7**

Read the scenario below and answer the questions.

7.1 You are part of a team of software developers and analysts tasked with upgrading an existing information system called iTaxIT due to changes in government legislation. The end users of the system also requested the inclusion of a hypertext editor within the system. You need to test whether the system meets the users' requirements. Successful testing will result in successful implementation.

7.1.1 Identify the type of system maintenance that must be done. (1)

7.1.2 Motivate the answer to QUESTION 7.1.1. (2)

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7.1.3 List THREE types of tests that your team can perform. (3)

7.1.4 Name ONE type of test that your team cannot perform. (1)

7.1.5 Motivate your answer to QUESTION 7.1.4. (1)

7.1.6 Differentiate between a *test plan* and a *test case*. (2)

7.1.7 Give THREE reasons for reviewing a project after implementation. (3)

7.1.8 Name THREE test deliverables that will be delivered as part of the entire testing process. (3)

7.2 Read the excerpt below and answer the questions.

**PICK AND PAY SUCCESSFULLY TRIALS BITCOIN AS PAYMENT**




Pick n Pay, one of South Africa's largest retailers, has successfully trialled letting customers pay for groceries using the cryptocurrency Bitcoin.

The retailer, together with tech companies Electrum and Luno, piloted letting staff pay for groceries via Bitcoin at its head office in Cape Town.

'We designed a process from user registration, tender acceptance at point of sale to funds authorisation, currency exchange in real time during the transaction and instant payment settlement,' said Richard van Rensburg, Pick n Pay's deputy CEO.

The pilot programme was limited to one store and is no longer active. Bitcoin is a digital currency for secure payments and storing money that does not require a banking institution.

[Adapted from: *http://www.fin24.com/Tech/News/pick-n-pay-successfully-trials-bitcoin-as-payment-20170919*]

7.2.1 Explain what a pilot implementation entails. (2 × 2) (4)

7.2.2 Give ONE reason why the retail supermarket used a pilot-adoption policy/approach. (2)

7.2.3 Discuss how a big-bang approach would have been different if Pick n Pay did NOT use the pilot-adoption strategy. (3 × 2) (6)

7.3 Define the term *go-live*. (1)

7.4 List THREE tasks that must be done before reviewing an information system. (3)

**[32]**

## QUESTION 8

8.1 Explain what an *autonomous robot* is. (2)

8.2 Explain THREE components of an *expert system*. (3 × 2) (6)

8.3 List THREE programming languages used in the field of artificial intelligence. (3)

8.4 Name THREE practical applications of *fuzzy logic*. (3)
**[14]**

**TOTAL SECTION B: 160**
**GRAND TOTAL: 200**

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